

5 (a) (i) sum of e.m.f.'s = sum of p.d.'s around a loop/circuit B1 [1]

(ii) energy B1 [1]

(b) (i) $2.0 = I \times (4.0 + 2.5 + 0.5)$ C1
 $I = 0.286 \text{ A}$ (allow 2 s.f.) A1 [2]
(If total resistance is not 7Ω , 0/2 marks)

(ii) $R = [0.90 / 1.0] \times 4 (= 3.6)$ C1
 $V = I R = 0.286 \times 3.6 = 1.03 \text{ V}$ A1 [2]
(If factor of 0.9 not used, then 0/2 marks)

(iii) $E = 1.03 \text{ V}$ A1 [1]

(iv) either no current through cell B
or p.d. across r is zero B1 [1]

6 (a) (i) coherence: constant phase difference M1